

Disorders of Metabolism and Endocrine Function and the Impact on Oral Health

OMFS 513: Integrated Biomedical and Clinical Sciences

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Mission: To advance the science and art of health through education, service, scholarship and social accountability

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Learning Outcomes & Competency Statements

Learning Outcomes

1. Discuss the clinical and oral manifestations of disease associated with selected portions of the endocrine system.
2. Discuss the dental implications of treating patients with disorders of the endocrine system.

Competency statements met

1.3, 2.1, 3.1, 4.4, and 5.1

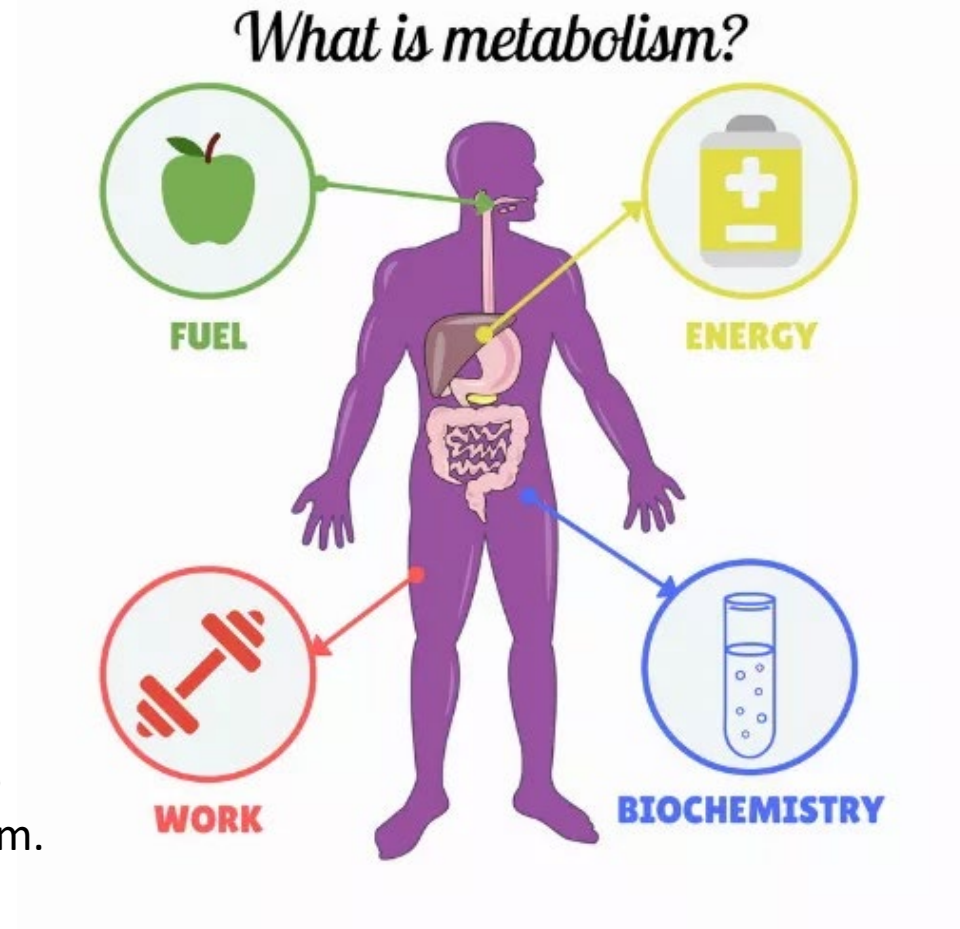
Big Picture

1. Review the function of the hormones secreted from the **thyroid gland, adrenal glands, and parathyroid glands**.
2. The role of the Hypothalamic-Pituitary-Axis (HPA)
3. How the pathology of hyper- and hyposecretion of the hormones affects systemic health. Recognize the clinical signs and symptoms.
4. How does the disease of various endocrine diseases play a role in dental treatment.
5. **Review Chapters 15 & 16** in the *Dental Management of the Medically Compromised Patient* textbook.

What is metabolism?

“**Metabolism**- the sum of the chemical reactions that take place within each cell of a living organism and that provide energy for vital processes and for synthesizing new organic material.”

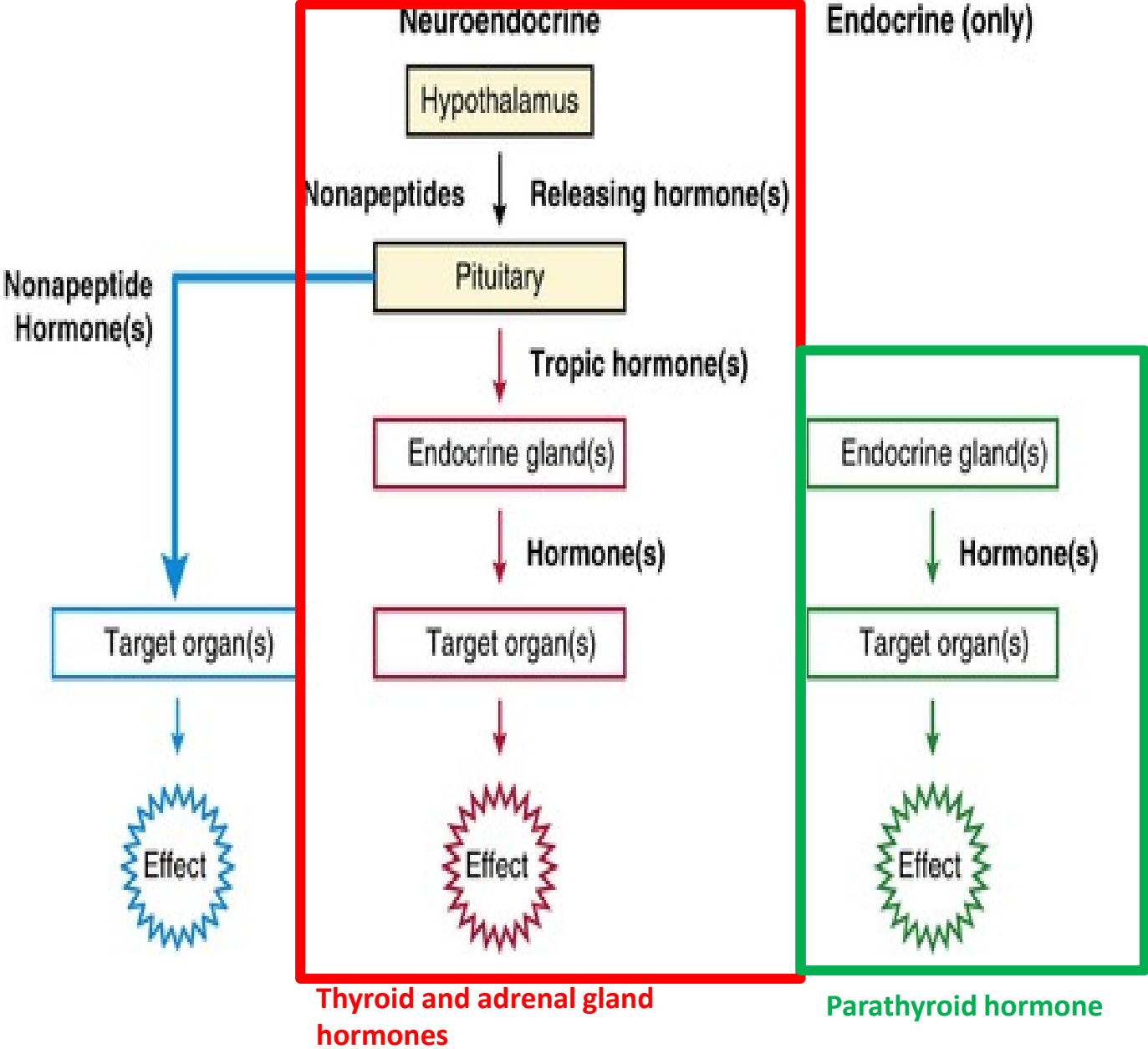
Kornberg, Hans. "metabolism". Encyclopedia Britannica, 16 Nov. 2021, <https://www.britannica.com/science/metabolism>. Accessed 4 May 2022.



Role of Hormones in Metabolism

- Hormones aid in the regulation of metabolic activity
 - Hormones control the function of the body's organs.
 - Influences the way the body stores and uses energy
 - Controls fluid, sugar, and salt levels
 - Provide communication between the brain, organ systems, and tissues in the body.

Endocrine System Subdivisions



Thyroid Gland

“Butterfly-shaped” gland in the midline of the neck

Aids in cellular turnover, growth in children, and energy/metabolism in adults.

Hormones of Thyroid

1. Thyroid releasing hormone

- Released from hypothalamus, acts on pituitary gland

2. Thyroid stimulating hormone

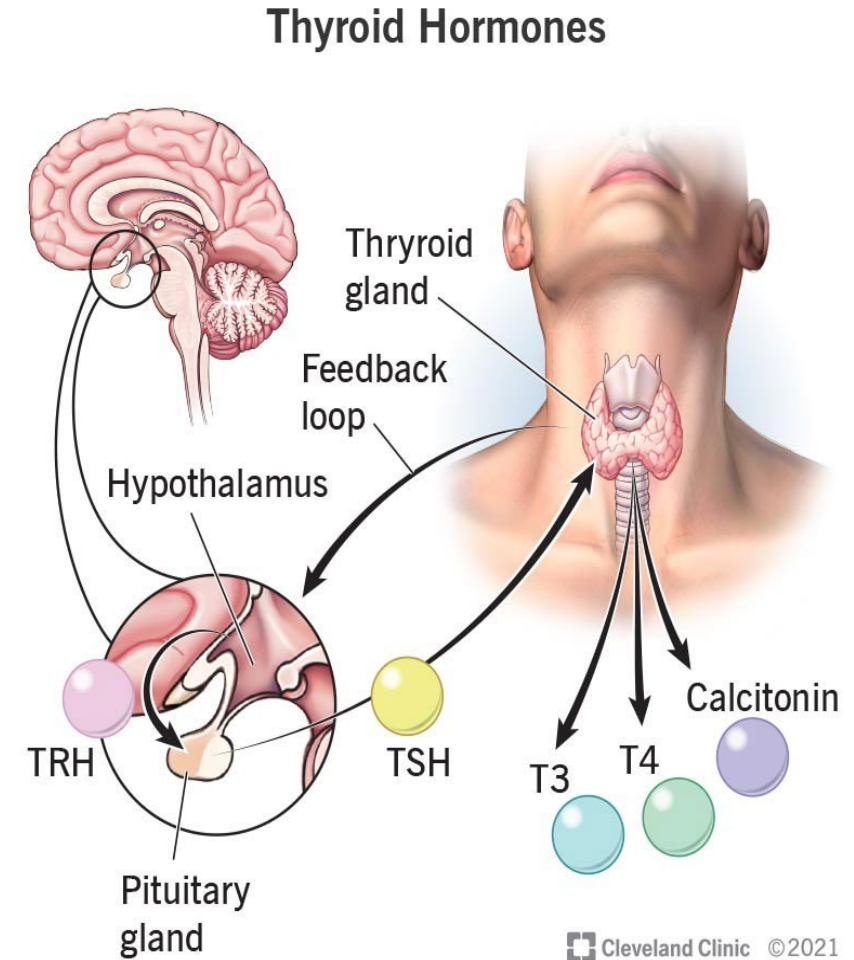
- Acts on thyroid gland to aid in the release of T3 and T4

3. Triiodothyronine (T3) and Thyroxine (T4)

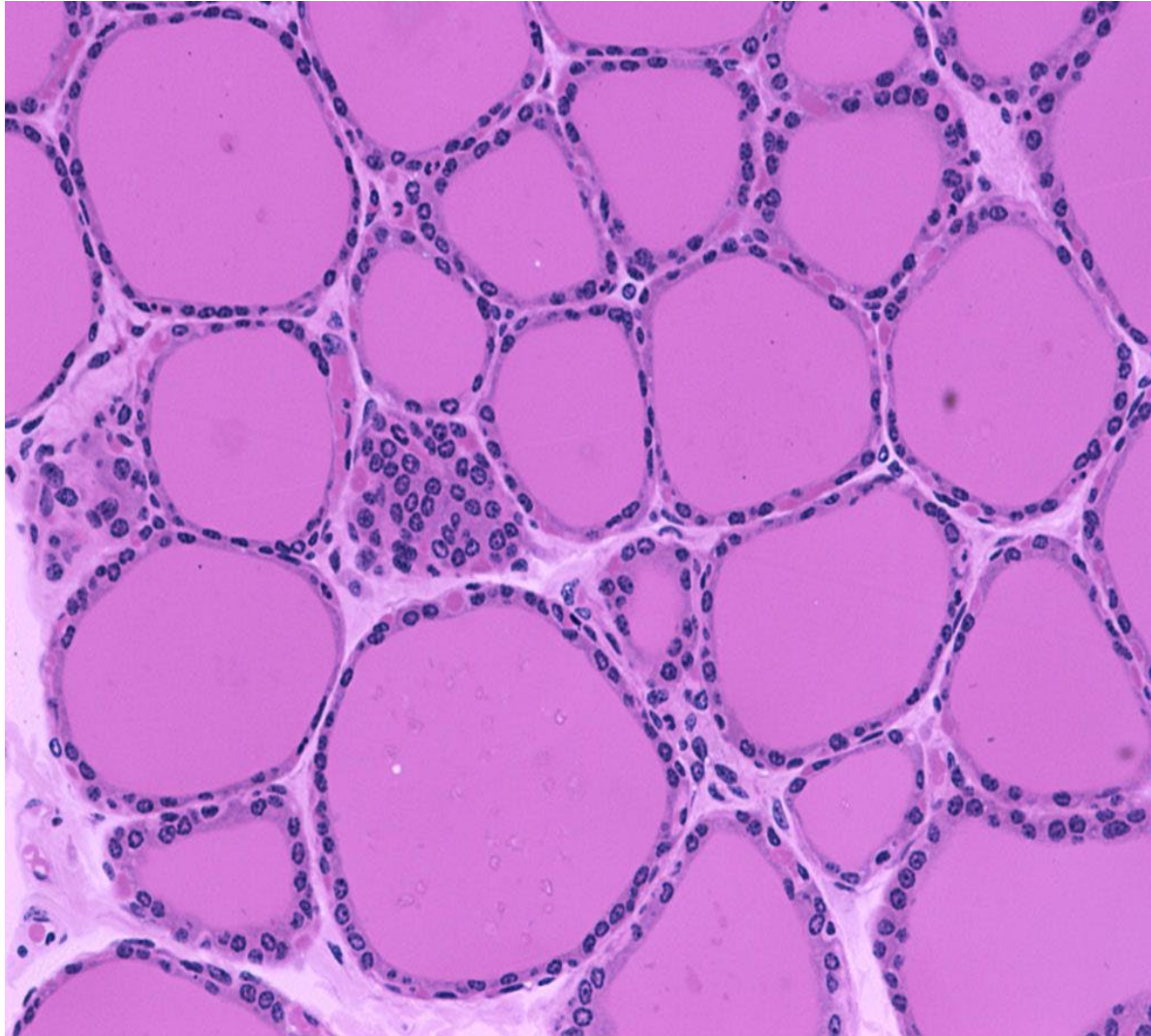
- Regulates weight, metabolism, energy, skin, hair, and nail growth.

4. Calcitonin

- Promotes Ca^{2+} retention in bones
- Decreases Ca^{2+} levels in blood



Thyroid Gland Histology



<https://www.proteinatlas.org/learn/dictionary/normal/thyroid+gland/detail+1/magnification+1>

Disorders of the Thyroid Gland

- Goiters
- Hyperthyroidism
- Hypothyroidism
- Thyroiditis – Hashimoto's Disease
- Neoplasm



www.endocrine.org/patient-engagement/endocrine-library/goiter

Goiters

- Irregular growth of the thyroid gland
- Etiology
 - Endemic
 - Geographic areas where iodine deficiency exist
 - Physiologic Demand
 - Occurs during puberty when the body has an increase physiologic demand for T4
 - Pregnancy
 - Drug/Food Consumption
 - Excessive ingestion of food or medication that interfere with thyroid hormone synthesis
 - Pathologic Disease

Goiters

- Classification by Texture/Appearance
 - Simple (Diffuse) – smooth texture
 - Nodular
 - Multinodular
 - Classifications by Function
 - Toxic (Functional) – excess thyroid hormone secretion
 - Non-Toxic (Non-functional) – normal or decreased thyroid hormone secretion
- } “lumpy” texture

Hyperthyroidism

“Thyrotoxicosis”

Causes:

- Ectopic thyroid tissue
- **Simple “Diffuse” hyperplasia –Graves disease**
- Ingestion of excess thyroid hormone
- Hyperfunctioning multinodular goiter “Toxic goiter”
- Hyperfunctional adenoma “Toxic adenoma”
- Pituitary disease- involves anterior portion of thyroid (Secondary disease)

Grave's Disease

- **Thyroid-stimulating immunoglobulins bind to receptors on thyroid follicles**
- Diffuse hypertrophy and hyperplasia of follicular epithelial cells
- **Elevated T_4 and T_3 levels; decreased TSH levels**
- Clinical Features:
 - 20–40-year-old
 - Female predilection
 - Exophthalmos, vision loss, pretibial myxedema
 - Coma and death
 - Rapid heart rate
 - Weight loss
 - Heat intolerance
 - Irritability
 - Changes of skin/hair/nails



www.thyroidcancer.com/blog/how-to-treat-graves-disease-top-3-options

Oral Manifestations of Hyperthyroidism

- Children
 - Rapid development of jaws and teeth
 - Premature loss of deciduous teeth
 - Early eruption of succedaneous teeth
- Adults
 - Posterolateral tongue enlargement
 - Lingual thyroid
 - Midline of posterior tongue
 - Xerostomia
 - Dental caries

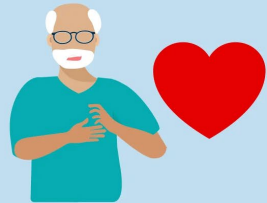
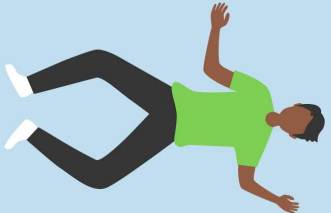
Hyperthyroidism Management

- Antithyroid medications
 - Propylthiouracil
 - methimazole
- Radioactive iodine – ^{131}I
- Surgery

Hyperthyroidism - Thyrotoxicosis

- **Thyroid Storm (Thyrotoxic Crisis)**
 - Abrupt onset of severe hyperthyroidism
 - Acute elevation of catecholamine levels during stress
 - Untreated patients die of cardiac arrhythmias
 - **Emergency**
 - Provide antithyroid medication
 - Propranolol
 - Keep patient cool with cold towels.
 - Administer IV hydrocortisone or oral dexamethasone
 - Hypertonic glucose

Symptoms of a Thyroid Storm

 High fever.	 Agitation.	 Delirium.
 Congestive heart failure.	 Loss of consciousness.	

 **Seek immediate medical care if you're experiencing these symptoms.**

Hypothyroidism

- **Primary**
 1. Surgical removal or radiation
 2. **Hashimoto's thyroiditis**
 3. Iodine insufficiency
 4. Drug-induced thyroid inhibition
 5. Congenital disease - **Cretinism**
- **Secondary**
 - Pituitary disease
- **Tertiary**
 - Disease involving the hypothalamus

Hypothyroidism

- **Myxedema**

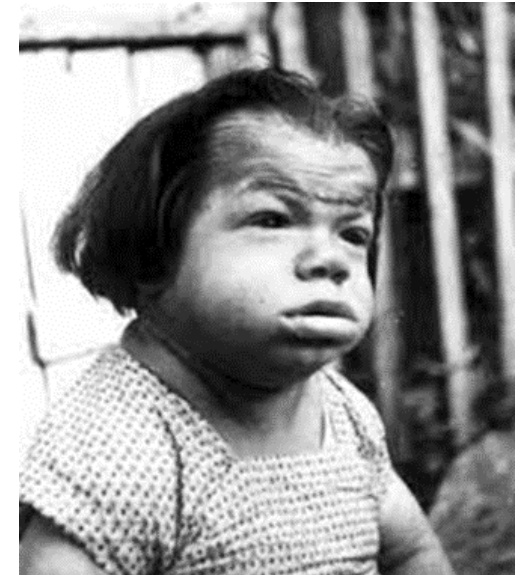
- Occurs in older children and adults
- Affected patients present with changes in personality (apathy), mental sluggishness, cold intolerance, and obesity.
- Edema in the soft tissues due to a build-up of mucopolysaccharide
- Enlargement of the heart or heart failure



Myxedema

- **Cretinism**

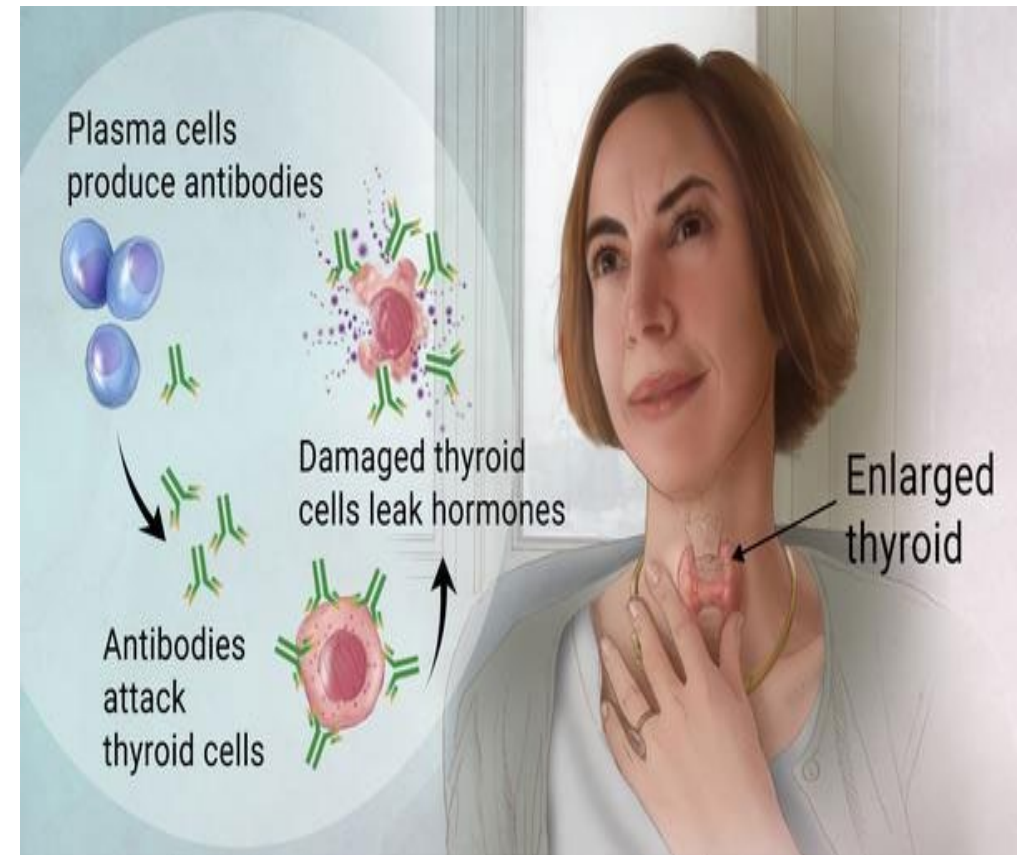
- Hypothyroidism in infants or children.
- Impaired skeletal and CNS developmental abnormalities.
- Facial features: short, coarse features



Cretinism

Hashimoto's Thyroiditis

- Autoimmune disease
- Antithyroid antibodies
- Female predilection
- 45-65 years of age
- Diffuse, firm, and symmetrically enlarged, painless thyroid goiter



Sources: Mayo Clinic

Oral Manifestations of Hypothyroidism

- Macroglossia
- Thickening of the lips
- Salivary gland enlargement
- Xerostomia
- Compromised periodontium
- Delay in tooth eruption in children
- Dysgeusia
- Burning mouth syndrome

Laboratory Testing

- **Hyperthyroidism**

 T_3 and T_4

Normal or  TSH

- **Hypothyroidism**

 T_3 and T_4

 TSH (slightly or normal)

Table 16.3

Thyroid Carcinomas

- <1% of cancer related deaths
- **Ionizing radiation most important risk factor**
- <1% of solitary thyroid nodules
- Types
 - Benign (adenomas)
 - Malignant
 - Medullary thyroid carcinoma – Multiple Endocrine Neoplasia Syndrome II

Thyroid-Possible Dental Complications

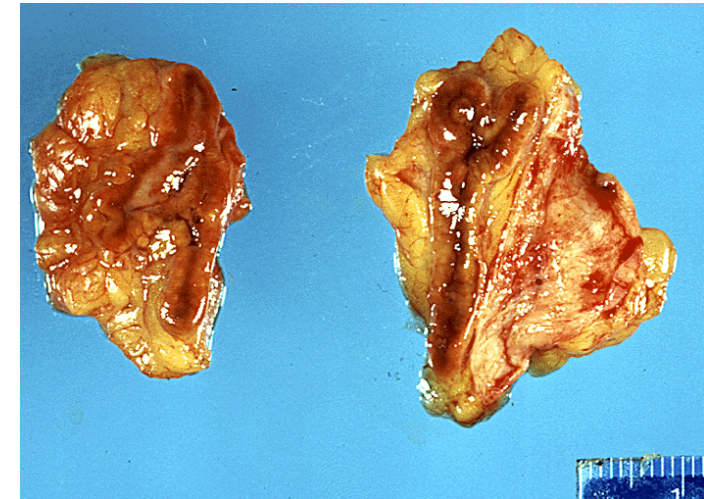
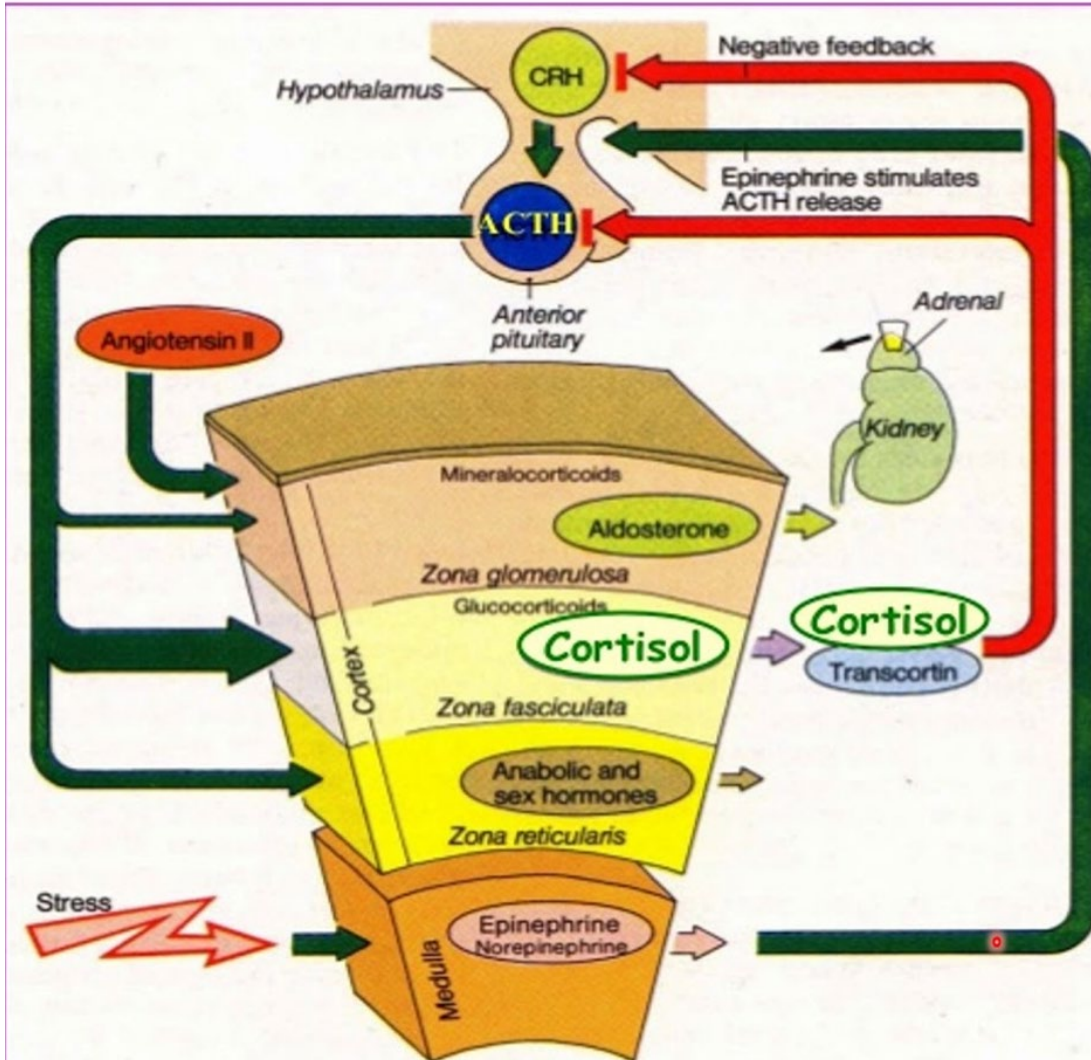
- **Hyperthyroidism**

- Adverse reactions with epinephrine
- High anxiety
- Bleeding risk (rare)
- Monitor blood pressure
- Cardiac arrhythmias
- Congestive heart failure
- **Increased risk of thyroid storm**
 - **Precipitated by infections and surgery**
- Aspirin and NSAID precautions

- **Hypothyroidism (poorly controlled)**

- Exaggerated response to CNS depressants
- Myxedematous coma due to CNS depressants, infection, or surgery
- Poor wound healing

Adrenal Cortex



Mineralocorticoid Excess (Hyperaldosteronism)

- Sodium retention and potassium excretion
 - Hypertension and hypokalemia
- Primary Aldosteronism – Autonomous overproduction of aldosterone → suppression of renin-angiotensin-aldosterone system.
 - Hyperplasia (15%)
 - Neoplasm (80%)
- Secondary Aldosteronism - Increase amount of aldosterone due to activation of the renin-angiotensin-aldosterone system.
 - Chronic heart failure, decreased renal perfusion, hypoalbuminemia

What does cortisol do in the body?

- Aids in state of stress, anxiety, infection, trauma, and illness
- Metabolism
 - Lipolysis
 - Proteolysis
 - Gluconeogenesis
 - Increased blood pressure
- Catecholamine release
- Digestion
- Anti-inflammatory
- Cardiovascular function

Cushing Disease/Syndrome (Cortisol Excess)

Clinical Presentation

Weight gain	Impaired/delayed wound healing
Abdominal striae	Psychosis
Hypertension	Insomnia
Glucose intolerance	Peptic ulceration
Hirsutism	Growth suppression
Heart failure	Suppressed immune response
Osteoporosis/bone fracture	Muscle weakness

Cushing's Disease/Syndrome Clinical Presentation



Fullness of the face-Moon facies
Striae on the axillary/pectoral region.



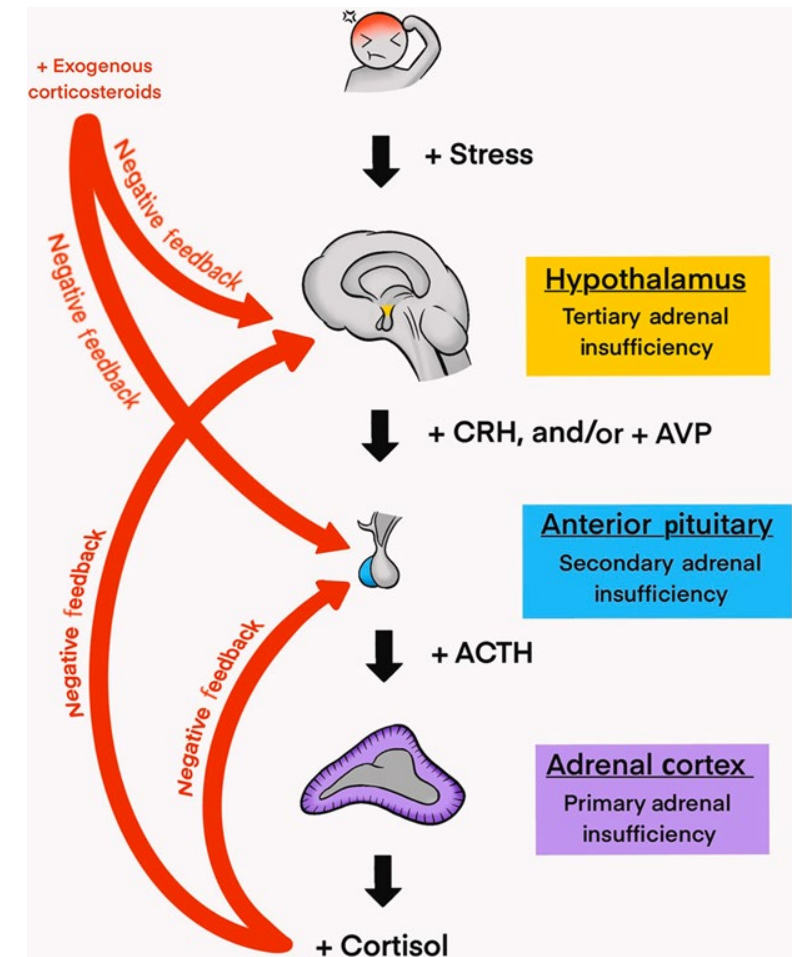
Fat deposition between the shoulder blades "buffalo hump"



Fat deposition in the abdominal area and striae on the skin

Adrenal Insufficiency - Hypoadrenalism

- Insufficient levels of adrenal hormone in circulation
- Primary-**Addison's Disease**
 - Destruction of the adrenal cortex
 - Acute: Emergency
 - Chronic
- Secondary-Disease of the pituitary gland
- Tertiary-Impaired function of the hypothalamus
 - Seen with exogenous use of steroids



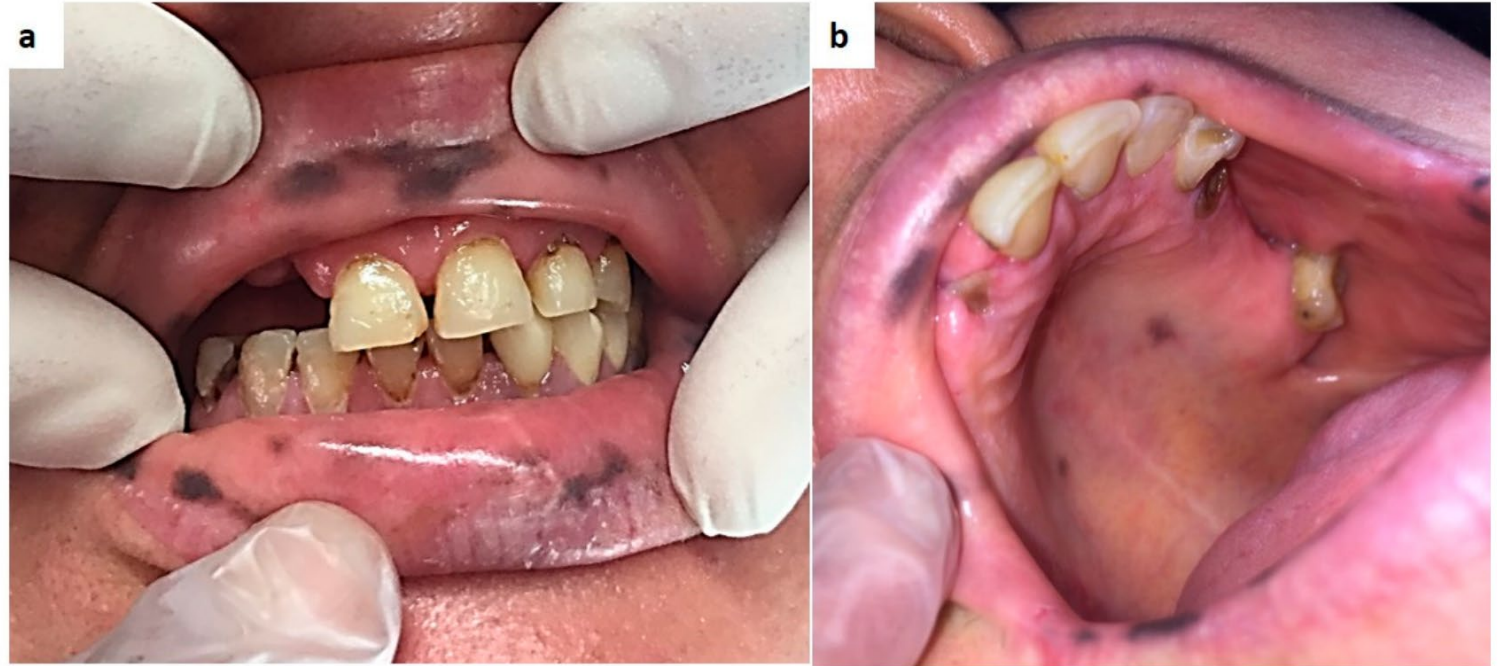
Chronic Adrenocortical Insufficiency

- **Addison's Disease**
- Progressive destruction of adrenal cortex
 - 90% must be destroyed before manifestations
- Autoimmune adrenalitis (60-70%)
 - Irregular shrunken glands
- Infections
 - Architecture destroyed by granulomatous inflammation
- Metastatic neoplasms

Chronic Adrenocortical Insufficiency – Addison's Disease

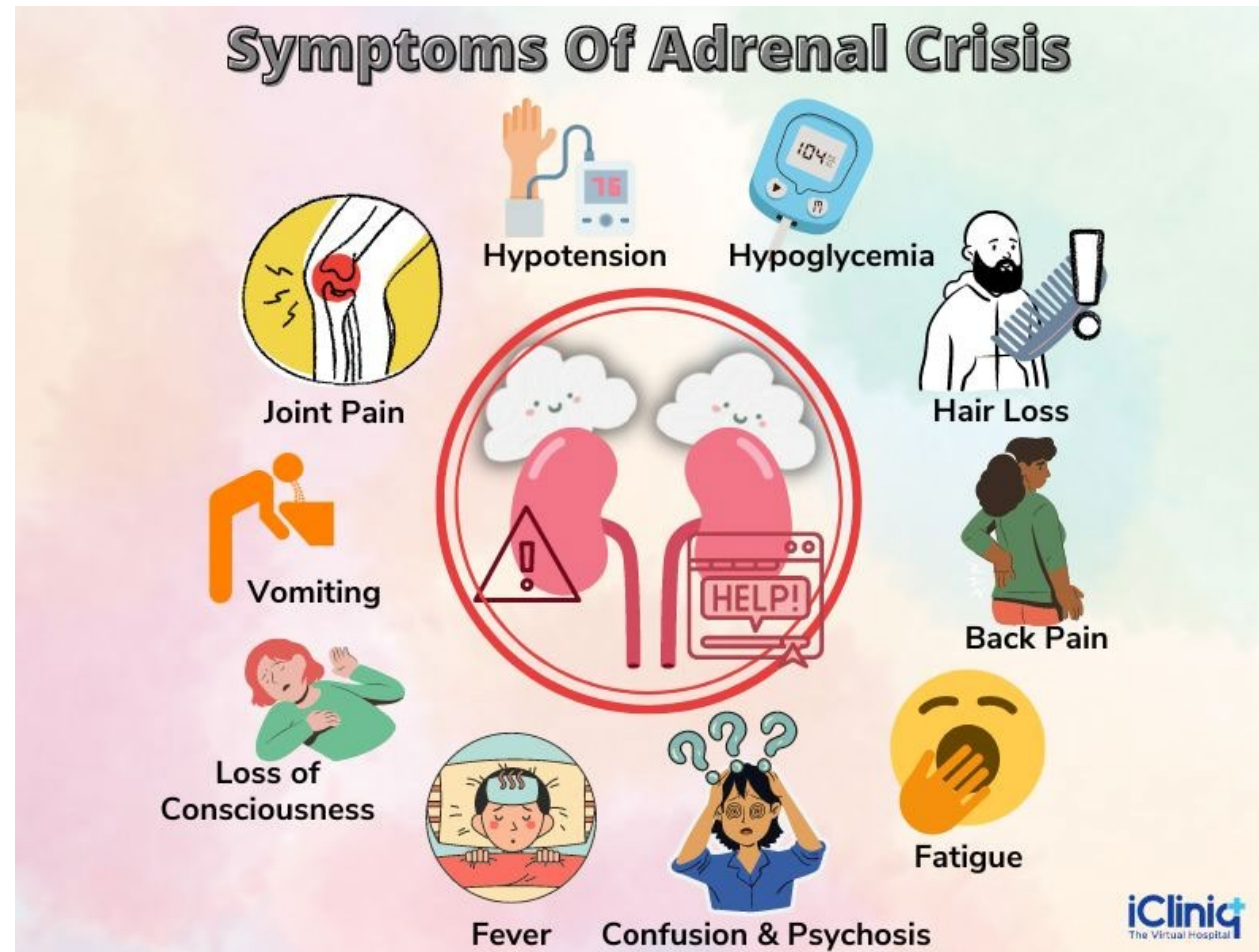
- **Progressive weakness and easy fatigability**
- GI symptoms: anorexia, nausea, vomiting, diarrhea
- **Hyperpigmentation of skin and oral mucosa**
- Decreased heart size from chronic hypovolemia
- Adrenal crisis precipitated by stresses
 - Vomiting, abdominal pain, hypotension, coma, vascular collapse
 - Require corticosteroids to prevent death

Chronic Adrenocortical Insufficiency – Addison's Disease



Oral and Maxillo-Facial Manifestations of Systemic Diseases: An Overview. Medicina 2021, 57, 271.

Adrenal Crisis



Adrenal Gland-Dental Complications

- Bone and tissue healing
- Candidiasis
- Exogenous steroid drug-drug interactions
- **Adrenal crisis**



Management of Dental Treatment with Adrenal Insufficiency

Table 5.3

- Primary Adrenal Insufficiency
 - No additional steroids required for routine procedures
 - Base line dosage for minor surgical procedures is 25mg of hydrocortisone preoperatively
 - The dosage will increase for more moderate and major procedures
- Secondary
 - Daily therapeutic does is suggested for all surgical procedures

Management of Adrenal Crisis

1. Place patient in a supine position
2. IV injection of glucocorticoids
3. Keep the patient cool with ice packs
4. Monitor blood pressure
5. Call EMS

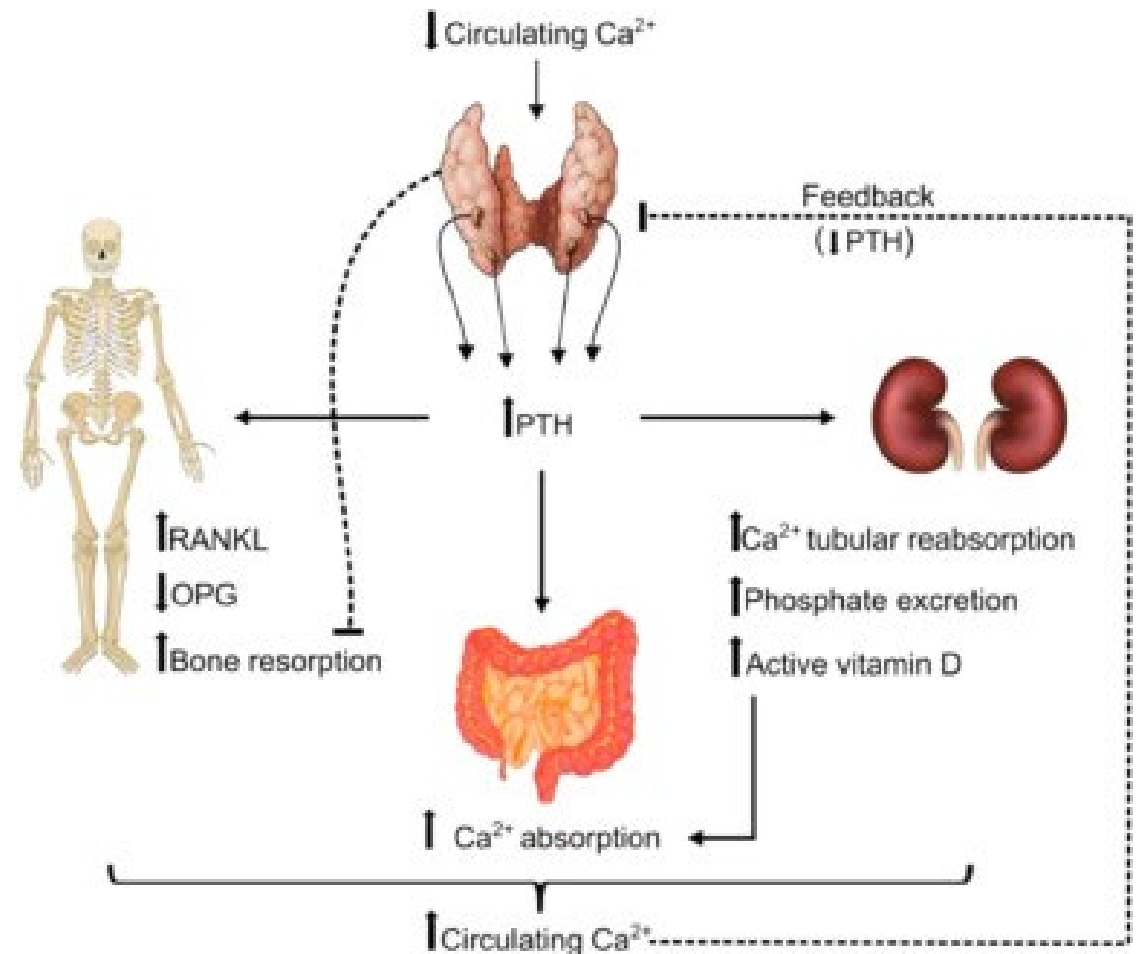
Adrenal Medulla

- Most important diseases are neoplasms
 - **Pheochromocytoma**
 - 10% Rule
 - Associated with Multiple Endocrine Neoplasia Type II Syndrome



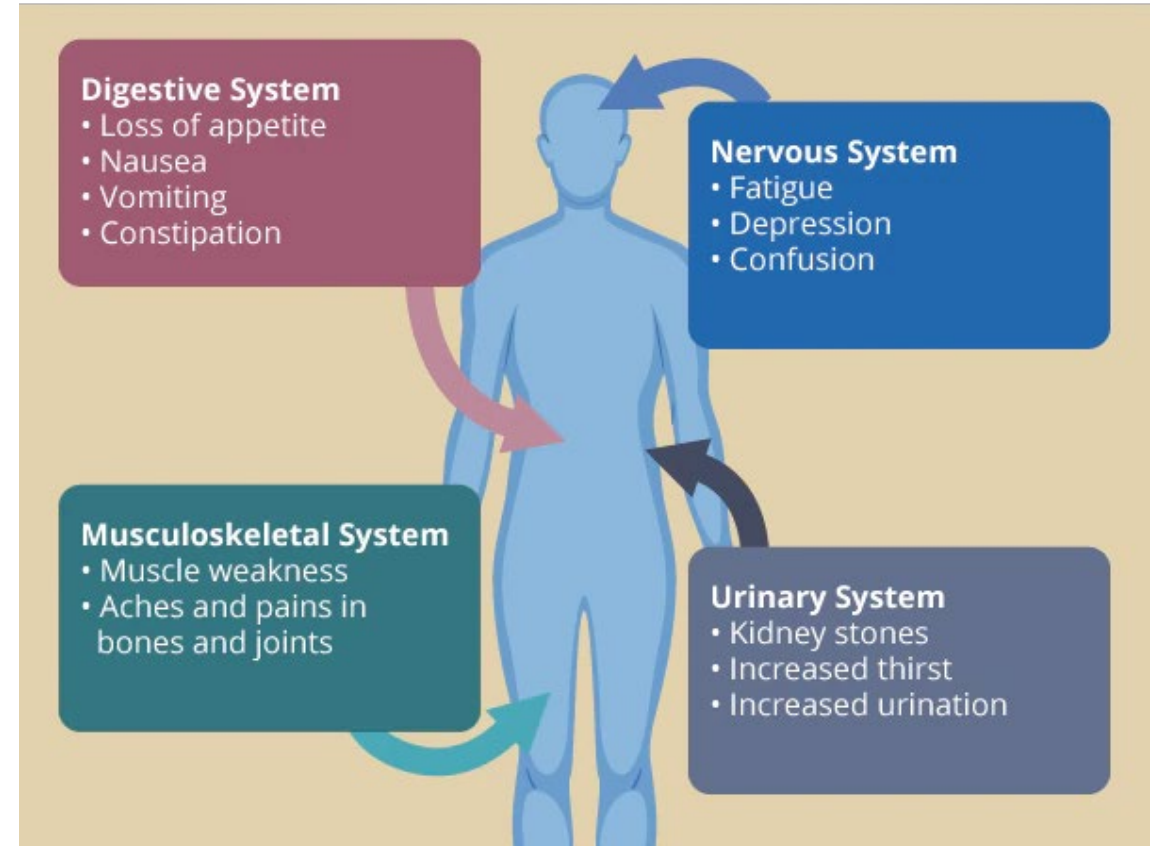
Parathyroid Glands

- Found within the thyroid gland tissue
- Activity controlled by level of free calcium
- Parathyroid hormone (PTH):
 - Activates osteoclasts
 - Increases renal reabsorption of calcium
 - Increases conversion of Vitamin D to dihydroxy form in kidneys
 - Increases urinary phosphate excretion
 - Augments GI Calcium absorption



Primary Hyperparathyroidism

- Adenoma or hyperplasia
- Parathyroid carcinoma
- Most common patients
 - Middle-aged women
- To help you remember these clinical features:
 - **“Stones, bones, abdominal groans, and psychic moans”.**



Secondary Hyperparathyroidism

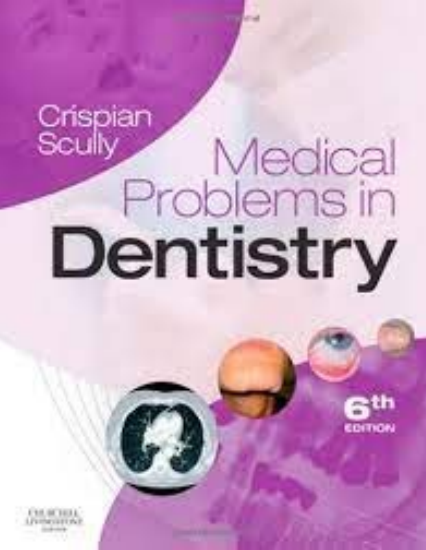
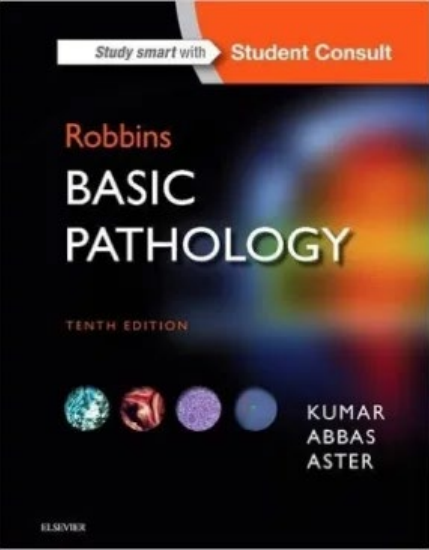
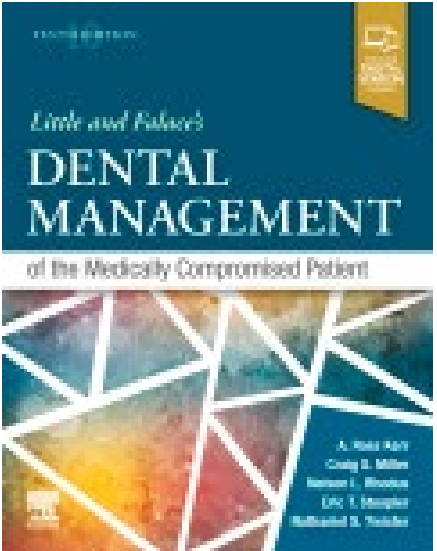
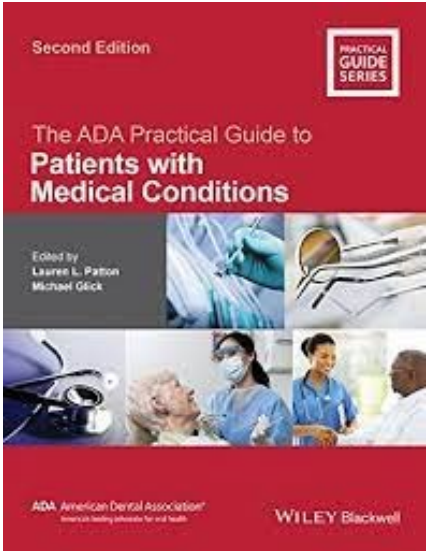
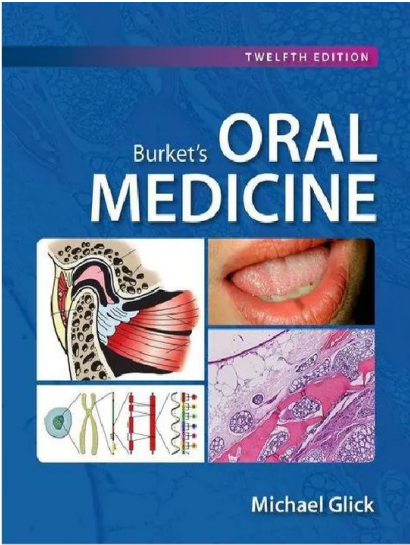
- Caused by any condition associated with chronic depression of serum calcium
- **Renal failure most common cause**
- Inadequate dietary intake of vit D deficiency
- Parathyroid glands are hyperplastic
 - Serum calcium levels near normal due to compensatory increase in parathyroid hormone.
- **Bone changes (Brown's tumors) and metastatic calcification**



Hypoparathyroidism

- Cause
 - Surgical removal or primary atrophy (decrease in size)
- Hypocalcemia: Tingling, neuromuscular irritability, occasional seizures.
 - **Chvostek's sign- twitching or contracture of facial muscle produced when tapping the facial nerve at a specific point on the face.**

Resources



FULL TEXT ARTICLE
Update on Management of the Oral and Maxillofacial Surgery Patient on Corticosteroids

Michael W. Clark DDS
Oral and Maxillofacial Surgery Clinics, 2022-02-01, Volume 34, Issue 1, Pages 115-120, Copyright © 2021 Elsevier Inc.

Corticosteroids have been the cornerstone for treatment of many inflammatory and immune disorders with these beneficial effects well recognized by the medical community. It also possesses many undesirable clinical adverse effects that can occur within 2 weeks of use. Moreover, in the past decade, chronic users of corticosteroids have been linked to delayed (retarded) and high interoperative outcomes with some patients requiring effective antiepileptic medications to counteract feature persistence. Additionally, two case reports have implicated daily prednisone use to cause osteonecrosis of the mandible. This chapter highlights current adrenal suppression classifications, pathophysiology, drug interactions, and perioperative surgical and anesthesia management.

Key points

- Corticosteroids have beneficial therapeutic properties but it also has a wide range of adverse clinical side effects.
- Most clinicians consider 3 weeks of continuous use of 20 mg of prednisone or its equivalent or higher to cause tertiary adrenal insufficiency and this phenomenon will likely cease after 12 months of drug discontinuation.
- Good perioperative pain control, especially during the postoperative phase, is crucial to decrease cortisol demand.
- High-quality randomized control trials are needed to determine if perioperative "stress dose" steroid is necessary for minor, moderate, and major surgeries.

Introduction

Corticosteroids have been used to treat a variety of anti-inflammatory and immunosuppression conditions such as arthritis (rheumatoid arthritis, psoriatic arthritis, rheumatoid arthritis), polymyalgia rheumatica, autoimmune disease (e.g., systemic lupus erythematosus, giant cell arteritis), chronic obstructive pulmonary disease (COPD), asthma, dermatologic lesions, inflammatory bowel disease, and mucocutaneous lesions, adjuvant chemotherapy regimens, and organ transplant recipients. It is estimated that 1.2% of the US population is on chronic steroids.¹ With an aging population, OMS clinicians will likely encounter these patients in their daily practice. Currently, physicians have been recommended to prescribe the lowest dose and the shortest course for targeted therapy to avoid numerous undesirable side effects, but unfortunately, some patients may require a lifelong treatment regimen. The purpose of this article is to highlight some of the current adrenal insufficiency classifications, significant drug-to-drug interactions, adverse effects, and current perioperative recommendations for patients taking long-term corticosteroids suffering from these chronic ailments.